Complete Nim-Words and Sharp Preperiods for the Four-Move Subtraction Family $\{5, 11, 15, c\}$

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Abstract

For the normal-play one-pile subtraction game with moves $\{5, 11, 15, c\}$, $c > 15$ odd, we determine the least eventual period and the least preperiod for every admissible c. All sufficiently large parameters $c \geq 59$ admit a single explicit infinite nim-word $B(01)^{(c-45)/2}0D(01)^{\infty}$ with two fixed digit blocks of lengths 44 and 29, giving the sharp formula $(T, p) = (c + 29, 2)$. The 21 remaining odd values are proved by finite recurrence certificates and include the exceptional pairs $(0, 26)$ at $c = 21$ and $(36, 20)$ at $c = 29$. Zhang previously reported the ultimate period-two pattern and exceptional branches as bounded computations for a wider family; those observations are not claimed as new. Historical novelty of the exact all-c statements remains unverified.

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1 Definition and principal result

For a finite subtraction set $S \subset \mathbb{Z}_{>0}$ let $g_S(n)$ be the unique sequence given by

$$g_S(n) = \operatorname{mex}\{g_S(n - s) : s \in S,\ s \leq n\}, \qquad \operatorname{mex} \varnothing = 0. \tag{1}$$

An eventual period $p \geq 1$ starts at index $T \geq 0$ if $g_S(n + p) = g_S(n)$ for every $n \geq T$. We call the smallest possible p the *least eventual period* and the earliest possible starting index the *least preperiod*. For a finite word W, write W^r for concatenation and W^{∞} for infinite repetition.

Let the following words be indexed from zero:

$$\begin{aligned} B &= \texttt{00000111110222213333020001011101022101330102}, & |B| &= 44, \\ D &= \texttt{12121030301010101010101210103}, & |D| &= 29. \end{aligned} \tag{2}$$

Theorem 1. *Let $c > 15$ be odd and put $S_c = \{5, 11, 15, c\}$.*

(i) *For every odd $c \geq 59$, the* complete *nim-word is*

$$\mathcal{G}_c = B\,(\texttt{01})^{(c-45)/2}\,\texttt{0}\,D\,(\texttt{01})^{\infty}, \tag{3}$$

with least *preperiod $T_c = c + 29$ and* least *eventual period $p_c = 2$.*

(ii) *For the remaining odd $c \in [17, 57]$, the least pairs (T_c, p_c) are those in Table 1. The entire infinite word for each case is specified by a finite mex-generated prefix of length T_c followed by the primitive period block of length p_c; the enclosed verifier and certificate table list those blocks explicitly.*

2 Proof of the uniform infinite family

We first record a finite periodic-word criterion.

c	T	p	c	T	p	c	T	p
17	46	2	19	52	2	21	0	26
23	58	2	25	54	2	27	56	2
29	36	20	31	80	2	33	62	2
35	64	2	37	66	2	39	68	2
41	96	2	43	98	2	45	74	2
47	76	2	49	78	2	51	80	2
53	82	2	55	84	2	57	86	2

Table 1: Finite certificate cases; all parameter values $c > 15$ below the uniform-word threshold 59.

Lemma 1 (Finite recurrence certificate)**.** *For a proposed infinite word f which is periodic from index T with period p, let $s_{\max} = \max S$. If f satisfies the mex recurrence (1) at every index $0 \le n \le T + p + s_{\max} - 1$, then f satisfies it at every index $n \ge 0$.*

Proof. Use strong induction. At $n \ge T + p + s_{\max}$ every predecessor of $n - p$ is legal and has index at least T. The lists of predecessor values for n and $n - p$ are therefore identical by the periodicity of f. Their mex values agree, and the recurrence at $n - p$ gives $f(n - p) = f(n)$, completing the induction. □

Lemma 2. *For the three-move set $S_0 = \{5, 11, 15\}$, one has $\mathcal{G}_{S_0} = B(\mathtt{01})^\infty$. In addition,*

$$\begin{cases} g_{S_0}(j) \neq 0 & \text{for all odd } j \ge 29, \\ g_{S_0}(j) \neq 1 & \text{for all even } j \ge 30. \end{cases} \tag{4}$$

Proof. Direct mex evaluation for $0 \le n \le 60$ agrees with $B(\mathtt{01})^\infty$ (a complete integer table is reproducible with the bundled script). Apply Lemma 1 with $T = 44, p = 2, s_{\max} = 15$. For the exclusions, check respectively the eight odd indices $29, 31, \ldots, 43$ and the seven even indices $30, 32, \ldots, 42$ of B; above 43 the suffix alternates. □

Proof of Theorem 1(i). Fix odd $c \ge 59$ and let f denote the word (3). For $n < c$, move c is forbidden and $f(n) = g_{S_0}(n)$, so Lemma 2 supplies the recurrence.

For $n = c + i$ with $0 \le i \le 28$, move c produces predecessor i, whose value is B_i. For each $s \in \{5, 11, 15\}$, when $i < s$ the index $c + i - s$ lies in $[44, c - 1]$ and has value $i \bmod 2$ (both c and s are odd). When $i \ge s$ its value is D_{i-s}, an earlier entry of the displayed block. Thus the 29 required equations are the c-independent finite checks

$$D_i = \operatorname{mex}\left(\{B_i\} \cup \left\{ \begin{cases} i \bmod 2, & i < s, \\ D_{i-s}, & i \ge s \end{cases} : s \in \{5, 11, 15\} \right\}\right), \quad 0 \le i \le 28. \tag{5}$$

All 29 integer equations are verified in the supplied `PATCH_29_MEX_TABLE.csv`, generated by `verify_no64.py`. Consequently the recurrence holds at every index in $[c, c + 28]$ for *all* eligible c, not merely for tested numerical instances.

Read directly from B, D, and the alternating blocks the global properties

$$f(j) \neq 0 \quad \text{for odd } j \ge 29, \qquad f(j) \neq 1 \quad \text{for even } j \ge 30. \tag{6}$$

Indeed, the finitely many exceptional indices in B are covered by (4); in D all even offsets i have nonzero values, and no odd offset has value 1. The other two parts of f alternate with parity.

For even $n \ge c + 29$, the predecessor indices $n - 5, n - 11, n - 15, n - c$ are all odd and at least 29. By (6), their values exclude zero, so their mex is $0 = f(n)$.

For odd $n \ge c + 29$, necessarily $n \ge c + 30$. All four predecessor indices are even and at least 30, so by (6) none has value 1. Furthermore, $n - 5$ is an even index at least $c + 25$. Both even indices $c + 25, c + 27$ have value 0 by (2), and all later even indices lie in the alternating suffix and have value 0. Hence $f(n - 5) = 0$, making the mex exactly $1 = f(n)$.

We have proved the recurrence at all indices. Uniqueness in (1) implies $f = \mathcal{G}_c$. Its suffix from $T = c + 29$ is nonconstant alternating and hence has least period 2. The preceding value $f(c + 28) = D_{28} = 3$, while all values from T onward belong to $\{0, 1\}$. Any period $q \ge 1$ starting before T would force $f(c + 28) = f(c + 28 + q)$, which is impossible. Therefore $T = c + 29$ is the least preperiod for *any* eventual period. □

3 Finite certificates for the remaining 21 values

Proof of Theorem 1(ii). For each row (c, T, p) of Table 1, the bundled verifier generates the true mex sequence through $N = T + p + c - 1$ and checks

$$g(n) = g(n + p) \qquad (T \le n \le T + c - 1). \tag{7}$$

Thus repeating the block $g(T), \ldots, g(T+p-1)$ matches the finite recursion through N, and Lemma 1 proves the infinite continuation. Each certificate also checks that the p-letter block has no proper divisor as a period. Therefore p is the primitive eventual period. Finally, for every $T > 0$, the exact integer check $g(T-1) \neq g(T-1+p)$ holds. Since every eventual period of a primitive p-periodic suffix is a multiple of p, an earlier start would force equality of these two values, a contradiction. When $T = 0$ minimality is immediate. The reproducible certificate table records the tested range, the boundary values and each primitive word. This proves both least parameters for every case. □

4 Scope, prior art, and a separate conjecture

The base three-move family is not a new result: Ho [3], Theorem 5, proves least preperiod $2a^2 - a - 1$ and eventual period 2 for odd $a \geq 5$ and $\{a, 2a+1, 3a\}$, giving $(T, p) = (44, 2)$ at $a = 5$. Lemma 2 independently verifies the explicit finite prefix and the parity exclusions needed here; it does not claim priority for the base-family periodicity.

Zhang [1], Remark 7, records bounded computations for the broader four-move family $\{a, 2a+1, 3a, c\}$ with odd a, c, $3 \leq a \leq 25$ and $c < 500$, including period 2 in generic cases, the pure-periodic branch $c = 4a+1$, and exceptional branches $c = (4i+2)a - 1$. In particular, $c = 21$ with $(T, p) = (0, 26)$ and $c = 29$ with $(T, p) = (36, 20)$ at $a = 5$ agree with those earlier observations; these numerical observations are not new. For the seven finite cases $c = 45, 47, \ldots, 57$, Table 1 also gives $T = c + 29$; the full displayed word (3), however, is asserted uniformly only for $c \geq 59$. A separate finite computation also agrees with that word at $c = 49$; this does not extend the analytical proof's parameter range.

Flammenkamp's original dissertation [4] was obtained and directly compared in relevant chapters, general theorems, and its numbered four-move families; no direct anticipation of the present statement was identified in that examination. The separately published expanded 1997 book edition (ISBN 3-932136-10-1) was not obtained or directly checked. Moriwaki [5], Theorems 2 and 4, require successive move gaps not to exceed the least move. For $\{5, 11, 15, c\}$ the gap $11 - 5 = 6 > 5$ rules out direct application. Theorem 4 also describes augmenting an eligible base set by moves $s_1 + s_r, 2s_1 + s_r, \ldots$; an eligible base subset of an all-odd target cannot produce this target as a nontrivial augmentation because its first new move would be even. This only rules out *direct application* of the cited theorems, not every possible consequence or other related work.

The prospective contribution is the complete nim-word for odd $c \geq 59$, an all-c proof for the fixed $a = 5$ slice, and sharp preperiod classification. Historical novelty remains **INCONCLUSIVE**; no priority claim or human peer-review claim is made. An external AI reviewer reported proof and computation PASS within the reviewed scope, but this is not a formal proof certificate or human peer review.

Numerical exploration suggests the *unproved* wider conjecture that for odd $a \geq 3$ and all sufficiently large odd $c > 3a$, the least parameters for $\{a, 2a+1, 3a, c\}$ are

$$T_c = c + 2a^2 - 4a - 1, \qquad p_c = 2.$$

Checks for $a = 3, 5, \ldots, 27$ do not establish an all-a theorem. The present theorem concerns *only* $a = 5$.

Disclosure. AI-assisted mathematical research, with reproducibility code. This is an unrefereed public preprint, version 1.0. Copyright © 2026 Ryutaro Yonezu; all rights reserved. Human peer review: not performed.

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